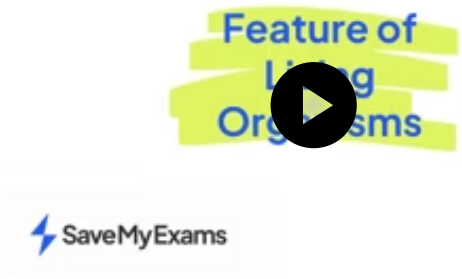
Features of Organisms

Exam code: 0610 & 0970

Download PDF

Written by: Lára Marie McIvor
Reviewed by: Lucy Kirkham
Updated on 23 September 2024



Did this video help you? Yes No

The Five Kingdoms

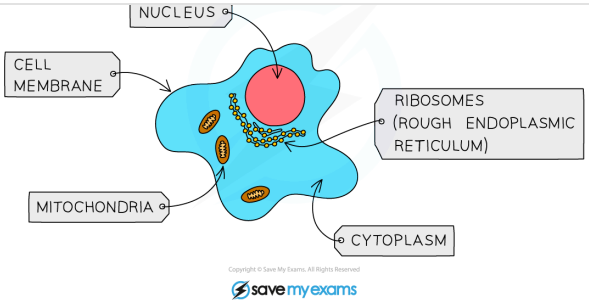
- The first division of living things in the classification system is to put them into one of **five kingdoms**. They are:
  - **Animals**
  - **Plants**
  - **Fungi**
  - **Protoctists**
  - **Prokaryotes**
- **Main features of all animals:**
  - they are multicellular
  - their cells contain a nucleus but no cell walls or chloroplasts
  - they feed on organic substances made by other living things

Help me understand

Stuck on something?

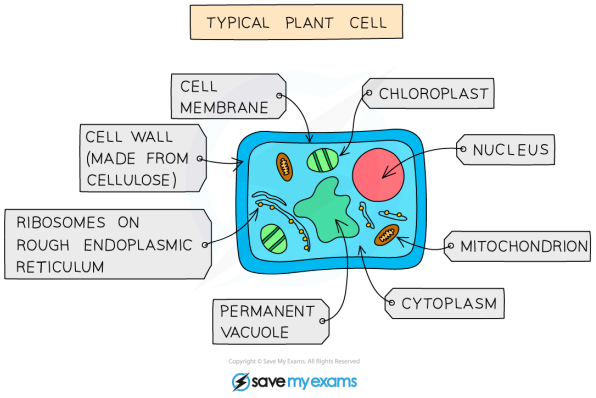
Ask questions and get help with exactly what you're revising.

Got it



A typical animal cell

- **Main features of all plants:**
  - they are multicellular
  - their cells contain a nucleus, chloroplasts and cellulose cell walls
  - they all feed by photosynthesis



A typical plant cell

The Animal Kingdom

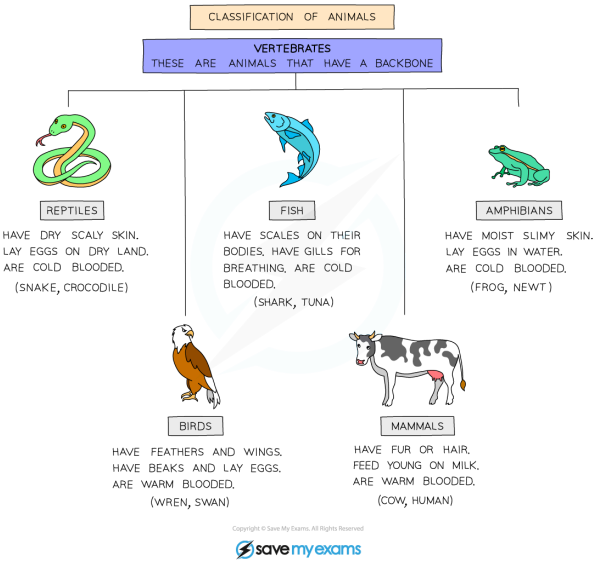
- Several main features are used to place organisms into groups within the animal kingdom

Vertebrates

- All vertebrates have a **backbone**
  - There are 5 classes of vertebrates

Vertebrate Table

	– YOUNG FEED ON MILK FROM MAMMARY GLANDS – EXTERNAL EARS (PINNA) VISIBLE – ENDOTHERMIC	
BIRDS	– SKIN COVERED IN FEATHERS – HAVE 2 LEGS AND 2 WINGS INSTEAD OF FORELIMBS – LAY EGGS WITH HARD SHELLS ON LAND – HAVE A BEAK – ENDOTHERMIC	PARROT, BLUE TIT, EAGLE
REPTILES	– DRY, FIXED SCALES ON SKIN – LAY EGGS WITH RUBBERY SHELLS ON LAND	SNAKE, TURTLE, IGUANA
AMPHIBIANS	– SMOOTH, MOIST SKIN – ADULTS USUALLY LIVE ON LAND (SO HAVE LUNGS), LARVAE LIVE IN WATER (SO HAVE GILLS) – LAY EGGS WITHOUT SHELLS IN WATER	FROG, TOAD, NEWT
FISH	– LOOSE, WET SCALES ON SKIN – GILLS TO BREATHE – LAY EGGS WITHOUT SHELLS IN WATER	FLOUNDER, GROUPER



Vertebrate classification

Invertebrates

- Invertebrates do not possess a backbone
- One of the morphological characteristics used to classify invertebrates is **whether they have legs or not**
- All invertebrates with **jointed legs** are part of the **arthropod** phylum
- They are classified further into the following groups:

Arthropod groups table

Arthropod classification

Examiner Tips and Tricks

Make sure that you are aware of the distinction between the **main features** of a group of organisms and the **defining features** of a group:

- An exam question might ask you to state the *main* features of a group of organisms; these are given above, and some of these main features are **shared between the groups**, e.g. all of the vertebrates have a backbone, and all of the arthropods have segmented bodies
- It is also possible that an exam question might ask you to give a *defining* feature of one of these groups, i.e. a feature that **sets a group apart from the other groups**, e.g. mammals have fur while the other vertebrates do not, and crustaceans have 2 pairs of antennae while this is not the case for the other groups of arthropods

Teacher only

New! Exclusive Teacher-Only Questions

- Fresh, expert-written questions
- Students can't access them
- Fair accurate tests - built fast

Test yourself Flashcards

Was this revision note helpful? Yes No

Previous Next

Course Study tools All topics

Resources

Learning Hub
Scholarship
Join
Past Papers
Solution Banks
Sitemap

Dashboard

Overview
Brand ambassador
Profile
Subscription
Settings
Log out

Company

About us
Exam Specificity
Content Quality
Promotions
Jobs
Terms
Privacy
Cookie Policy
Help and Support

Subjects

Biology
Chemistry
Physics
Maths
Geography
English Literature
Psychology
All Subjects

© Copyright 2015–2026 Save My Exams Ltd. All Rights Reserved.